Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In an atom, an electron is moving with a speed of $600\ \mathrm{m/s}$ with an accuracy of 0.005%. Certainity with which the position of the electron can be located is $\left(\mathrm{h} = 6.6 \times 10^{-34}\ \mathrm{kg\ m^2\ s^{-1}}\right.$, mass of electron, $\left.e_m = 9.1 \times 10^{-31}\ \mathrm{kg}\right)$

(A) $5.10 \times 10^{-3}\ \mathrm{m}$

(B) $1.92 \times 10^{-3}\ \mathrm{m}$

(C) $3.84 \times 10^{-3}\ \mathrm{m}$

(D) $1.52 \times 10^{-4}\ \mathrm{m}$

Q.2

The position of both, an electron and a helium atom is known within $1.0\ \mathrm{nm}$. Further the momentum of the electron is known within $5.0 \times 10^{-26}\ \mathrm{kg\ ms^{-1}}$. The minimum uncertainty in the measurement of the momentum of the helium atom is

(A) $50\ \mathrm{kg\ ms^{-1}}$

(B) $80\ \mathrm{kg\ ms^{-1}}$

(C) $8.0 \times 10^{-26}\ \mathrm{kg\ ms^{-1}}$

(D) $5.0 \times 10^{-26}\ \mathrm{kg\ ms^{-}}$

Q.3

At temperature T, the average kinetic energy of any particle is $\frac{3}{2}\mathrm{KT}$. The de Broglie wavelength follows the order:

(A) Visible photon $>$ Thermal neutron > Thermal electron

(B) Thermal proton $>$ Thermal electron Visible photon

(C) Thermal proton $>$ Visible photon $>$ Thermal electron

(D) Visible photon $>$ Thermal electron Thermal neutron

Q.4

The de Broglie wavelength of a car of mass $1000\ \mathrm{kg}$ and velocity $36\ \mathrm{km/hr}$ is :

(A) $6.626 \times 10^{-34}\ \mathrm{m}$

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(B) $6.626 \times 10^{-38}\ \mathrm{m}$

(C) $6.626 \times 10^{-31}\ \mathrm{m}$

(D) $6.626 \times 10^{-30}\ \mathrm{m}$

Q.5

According to Bohr's theory, the angular momentum of an electron in 5th orbit is

(A) $10\ \mathrm{h}/\pi$

(B) $2.5\ \mathrm{h}/\pi$

(C) $25\ \mathrm{h}/\pi$

(D) $1.0\ \mathrm{h}/\pi$

Q.6

If the de-Broglie wavelength of a particle of mass m is 100 times its velocity, then its value in terms of its mass (m) and Planck's constant (h) is

(A) $\frac{1}{10}\sqrt{\frac{m}{h}}$

(B) $10\sqrt{\frac{h}{m}}$

(C) $\frac{1}{10}\sqrt{\frac{h}{m}}$

(D) $10\sqrt{\frac{m}{h}}$

Q.7

Calculate the velocity of ejected electron from the metal surface when light of frequency $2 \times 10^{15}\ \mathrm{Hz}$ fall on the metal surface and the threshold frequency is $7 \times 10^{14}\ \mathrm{Hz}$ for metal ?

(A) 1.37×10^{6}

(B) 1.26×10^{6}

(C) 1.45×10^{7}

(D) 1.37×10^{7}

Q.8

Two fast moving particles X and Y are associated with de Broglie wavelengths $1\ \mathrm{nm}$ and $4\ \mathrm{nm}$ respectively. If mass of X is nine times the mass of Y, the ratio of kinetic energies of X and Y would be

(A) $3:1$

(B) $9:1$

(C) $5:12$

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(D) $16:9$

Q.9

Which of the following radiations has the highest wave number?

(A) X-rays

(B) Microwaves

(C) I.R. rays

(D) Radio waves

Q.10

If a proton and α -particle are accelerated through the same potential difference, the ratio of deBroglie wavelengths λ_{p} and λ_{α} is

(A) 3

(B) $2\sqrt{2}$

(C) 1

(D) 2

Q.11

The velocity of particle A is $0.1\ \mathrm{ms}^{-1}$ and that of particle B is $0.05\ \mathrm{ms}^{-1}$. If the mass of particle B is five times that of particle A, then the ratio of deBroglie wavelengths associated with the particles A and B is

(A) $2:5$

(B) $3:4$

(C) $6:4$

(D) $5:2$

Q.12

The threshold frequency of a metal is $1 \times 10^{15}\ \mathrm{s}^{-1}$. The ratio of the maximum kinetic energies of the photoelectrons when the metal is irradiated with radiations of frequencies $1.5 \times 10^{15}\ \mathrm{s}^{-1}$ and $2.0 \times 10^{15}\ \mathrm{s}^{-1}$ respectively would be

(A) $4:3$

(B) $1:2$

(C) $2:1$

(D) $3:4$

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Q.13

If the following matter waves travel with equal velocity, the longest wavelength is that of a/an.

(A) electron

(B) proton

(C) neutron

(D) α - particle

Q.14

If numerical value of mass and velocity are equal then de-Broglie wavelength in terms of KE is

(A) $\frac{\mathrm{mh}}{2\mathrm{KE}}$

(B) $\frac{\mathrm{vh}}{2\mathrm{KE}}$

(C) both are correct

(D) none is correct

Q.15

If uncertainty in position and velocity are equal then uncertainty in momentum will be

(A) $\frac{1}{2}\sqrt{\frac{\mathrm{mh}}{\pi}}$

(B) $\frac{1}{2}\sqrt{\frac{\mathrm{h}}{\pi \mathrm{m}}}$

(C) $\frac{\mathrm{H}}{4\pi \mathrm{m}}$

(D) $\frac{\mathrm{mh}}{4\pi}$

Q.16

When a certain metal was irradiated with light of frequency $3.2 \times 10^{16}\ \mathrm{Hz}$, photoelectrons emitted had twice the kinetic energy as did photoelectrons emitted when the same metal was irradiated with light of frequency $2.0 \times 10^{16}\ \mathrm{Hz}$. Hence, threshold frequency is

(A) $0.8 \times 10^{15}\ \mathrm{Hz}$

(B) $8.0 \times 10^{15}\ \mathrm{Hz}$

(C) $0.8 \times 10^{14}\ \mathrm{Hz}$

(D) $6.4 \times 10^{16}\ \mathrm{Hz}$

Q.17

The values of Planck's constant is $6.63 \times 10^{-34}\mathrm{Js}$. The velocity of light is $3.0 \times 10^{8}\ \mathrm{m\ s^{-1}}$. Which value

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is closest to
the wavelength in nanometres of a quantum of light with frequency of $8 \times 10^{15}\ \mathrm{s}^{-1}$?

(A) 5×10^{-18}

(B) 4×10^{1}

(C) 3×10^{7}

(D) 2×10^{-25}

Q.18

If uncertainty in position and momentum are equal, then uncertainty in velocity is :

(A) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

(B) $\sqrt{\frac{h}{2\pi}}$

(C) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$

(D) $\sqrt{\frac{h}{\pi}}$

Q.19

If the radius of first orbit of H atom is a_0, the de-Broglie wavelength of an electron in the third orbit is

(A) $4\pi a_0$

(B) $8\pi a_0$

(C) $6\pi a_0$

(D) $2\pi a_0$

Q.20

If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become

(A) one fourth

(B) half

(C) four times

(D) two times

Answers & Solutions

Q.1 Answer:

$1.92 \times 10^{-3}\ \mathrm{m}$

Solution:

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According to Heisenberg uncertainty principle. $\Delta x \cdot m\Delta \mathrm{v}=\frac{h}{4\pi}$

$\Delta x=\frac{h}{4\pi m\Delta \mathrm{v}}$

Here $\Delta \mathrm{v}=\frac{600\times 0.005}{100}=0.03$

So, $\Delta x=\frac{6.6\times 10^{-34}}{4\times 3.14\times 9.1\times 10^{-31}\times 0.03}$

$=1.92\times 10^{-3}$

Q.2 Answer:

$5.0\times 10^{-26}\ \mathrm{kg\ ms^{-}}$

Solution:

According to Heisenberg uncertainty principle,

$\Delta x\times \Delta p=\frac{h}{4\pi}$ **(which is constant).**

As Δx **for electron and helium atom is same, thus momentum of electron and helium will also be same, therefore the momentum of helium atom is equal to** $5\times 10^{-26}\ \mathrm{kg\ ms^{-1}}$.

Q.3 Answer:

Visible photon $>$ Thermal electron Thermal neutron

Solution:

Kinetic energy of any particle $=\frac{3}{2}\mathrm{KT}$

Also K.E. $=\frac{1}{2}\mathrm{mv}^2$

$\frac{1}{2}mv^2=\frac{3}{2}KT\Rightarrow v^2=\frac{3KT}{m}$

$\mathrm{v}=\sqrt{\frac{3\mathrm{KT}}{\mathrm{m}}}$

De-broglie wavelength $=\lambda=\frac{\mathrm{h}}{\mathrm{mv}}=\frac{\mathrm{h}}{\mathrm{m}\sqrt{\frac{3\mathrm{KT}}{\mathrm{m}}}}$

$\lambda=\frac{\mathrm{h}}{\sqrt{3\ \mathrm{KTm}}}\quad \lambda\propto\frac{1}{\sqrt{\mathrm{m}}}$

Mass of electron $<$ **mass of neutron** λ **(electron)** $>$ λ **(neutron)**

Q.4 Answer:

$6.626\times 10^{-38}\ \mathrm{m}$

Solution:

$\lambda=\frac{h}{mv}$

$h=6.6\times 10^{-34}\ \mathrm{J-s}$

$m=1000\ \mathrm{kg}$

$v=36\ \mathrm{km/hr}=\frac{36\times 10^3}{60\times 60}\ \mathrm{m/sec}=10\ \mathrm{m/sec}$

$\therefore\quad \lambda=\frac{6.6\times 10^{-34}}{10^3\times 10}=6.6\times 10^{-38}\ \mathrm{m}$

Q.5 Answer:

$2.5\ \mathrm{h}/\pi$

Solution:

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Angular momentum of an electron in n^{th} orbit is given by

$\mathrm{mvr} = \frac{\mathrm{nh}}{2\pi}$

For $\mathrm{n} = 5$, we have

Angular momentum of electron

$= \frac{5h}{2\pi} = \frac{2.5h}{\pi}$

Q.6 Answer:

$10\sqrt{\frac{h}{m}}$

Solution:

Let wavelength of particle be x

So, velocity $(v) = \frac{x}{100}$

$\lambda = \frac{\mathrm{h}}{\mathrm{mv}}; \mathrm{x} = \frac{\mathrm{h} \times 100}{\mathrm{m} \times \mathrm{x}}$

$\mathrm{x}^2 = 100\frac{\mathrm{h}}{\mathrm{m}}$ **or** $\mathrm{x} = 10\sqrt{\frac{\mathrm{h}}{\mathrm{m}}}$

Q.7 Answer:

1.37×10^6

Solution:

$\frac{1}{2}mv^2 = hv - hv_0$

$\Rightarrow \frac{1}{2}mv^2 = h\left(v - v_0\right)$

$\Rightarrow \quad v = \sqrt{\frac{2h}{m}\left(v - v_0\right)}$

$= \sqrt{\frac{2 \times 6.6 \times 10^{-34}}{9.1 \times 10^{-31}}\left(2 \times 10^{15} - 7 \times 10^{14}\right)}$

$= 1.37 \times 10^6\ \mathrm{ms}^1$

Q.8 Answer:

$16 : 9$

Solution:

de Broglie wavelength $\lambda = \frac{h}{mv}$

$\frac{\lambda_1}{\lambda_2} = \frac{m_2 v_2}{m_1 v_1}; \frac{1}{4} = \frac{1}{9} \times \frac{v_2}{v_1}$

$\frac{v_2}{v_1} = \frac{9}{4}; \frac{v_1}{v_2} = \frac{4}{9}$

$\mathrm{KE} = \frac{1}{2}mv^2$

$\frac{KE_1}{KE_2} = \frac{m_1}{m_2} \times \frac{v_1^2}{v_2^2} = \frac{9}{1} \times \left(\frac{4}{9}\right)^2 = \frac{16}{9}$

Q.9 Answer:

X-rays

Solution:

Since $v \propto \bar{v} \propto \frac{1}{\lambda}$, the order is X-Rays, I.R. Rays, Microwaves, Radiowaves

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Q.10 Answer:

$2\sqrt{2}$

Solution:

$\lambda_p = \frac{h}{\sqrt{2eVm_p}};$
$\lambda_{He^{2+}} = \frac{h}{\sqrt{2\times 2eVm_{He^{2+}}}} = \frac{h}{\sqrt{2\times 2eV\times 4\ m_p}}$
$\therefore \frac{\lambda_p}{\lambda_{He^{2+}}} = 2\sqrt{2}$

Q.11 Answer:

$5 : 2$

Solution:

Given, $v_A = 0.1\ ms^{-1}$ **and** $v_B = 0.05\ ms^{-1}$ **also,**
$m_B = 5\ m_A$
de-Broglie wavelength, $\lambda = \frac{h}{mv}$
$\therefore \frac{\lambda_A}{\lambda_B} = \frac{h/m_A v_A}{h/m_B V_B} = \frac{m_B V_B}{m_A v_A}$
$= \frac{5\ m_A\times 0.05}{m_A\times 0.1} = 5\times 0.5 = 2.5 = 5/2$
$\therefore \lambda_A : \lambda_B = 5 : 2$

Q.12 Answer:

$1 : 2$

Solution:

$\frac{(KE)_1}{(KE)_2} = \frac{v_1 - v_0}{v_2 - v_0}$
$= \frac{\left(1.5\times 10^{15} - 1\times 10^{15}\right)}{(2.0\times 10^{15} - 1\times 10^{15})} = \frac{1}{2}$

Q.13 Answer:

electron

Solution:

$\lambda = \frac{h}{mv}$ **λ will be large if m is small**

Q.14 Answer:

both are correct

Solution:

Given $m = v$
$\therefore \quad \lambda = \frac{h}{mv} = \frac{h}{v^2}$
Also $KE = \frac{1}{2}mv^2$
$\therefore \quad v^2 = \frac{2KE}{m}$
$\therefore \quad \lambda = \frac{h}{\left(\frac{2KE}{m}\right)} = \frac{mh}{2KE}$

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Q.15 Answer:

$\frac{1}{2}\sqrt{\frac{\mathrm{mh}}{\pi}}$

Solution:

$\Delta \mathrm{x}=\Delta \mathrm{v}$
$\Delta \mathrm{x} \cdot \Delta \mathrm{v}=\frac{\mathrm{h}}{4 \pi \mathrm{m}}$
$\Delta \mathrm{v}^2=\frac{\mathrm{h}}{4 \pi \mathrm{m}}$
$\therefore \quad \Delta v=\sqrt{\frac{h}{4 \pi m}}$
$\Delta \mathrm{p}=\mathrm{m} \Delta \mathrm{v}=\mathrm{m} \sqrt{\frac{\mathrm{h}}{4 \pi \mathrm{m}}}$
$=\frac{1}{2} \sqrt{\frac{\mathrm{hm}}{\pi}}$

Q.16 Answer:

$8.0 \times 10^{15} \mathrm{~Hz}$

Solution:

$h v=h v_0+(K E)$
$K E=h\left(v-v_0\right)$
$(\mathrm{KE})_1=\mathrm{h}\left(3.2 \times 10^{16}-\mathrm{v}_0\right)$
$(\mathrm{KE})_2=\mathrm{h}\left(2.0 \times 10^{16}-\mathrm{v}_0\right)$
Given $\quad(\mathrm{KE})_1=2(\mathrm{~K.E.})_2$
$\mathrm{H}\left(3.2 \times 10^{16}-\mathrm{v}_0\right)=2 \mathrm{~h}\left(2.0 \times 10^{16}-\mathrm{v}_0\right)$
$\therefore \mathrm{v}_0=8.0 \times 10^{15} \mathrm{~Hz}$

Q.17 Answer:

4×10^1

Solution:

$\mathrm{E}=\mathrm{hv}=\frac{\mathrm{ch}}{\lambda}$; **and** $v=\frac{\mathrm{c}}{\lambda}$
$8 \times 10^{15}=\frac{3.0 \times 10^8}{\lambda}$
$\therefore \lambda=\frac{3.0 \times 10^8}{8 \times 10^{15}}=0.37 \times 10^{-7}=37.5 \times 10^{-9} \mathrm{~m}=4 \times 10^1$

Q.18 Answer:

$\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

Solution:

We know $\Delta p . \Delta x \geq \frac{h}{4 \pi}$
since $\Delta p=\Delta x$ **(given)** $\therefore \Delta p \cdot \Delta p=\frac{h}{4 \pi}$
or $m \Delta v m \Delta v .=\frac{h}{4 \pi}[\therefore \Delta p=m \Delta v]$

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or $(\Delta v)^2 = \frac{h}{4\pi m^2}$

or $\Delta v = \sqrt{\frac{h}{4\pi m^2}} = \frac{1}{2m}\sqrt{\frac{h}{\pi}}$

Q.19 Answer:

$6\pi a_0$

Solution:

$r_n = a_0 n^2$

$r = a_0 \times (3)^2 = 9a_0$

$mvr = \frac{nh}{2\pi}; \quad mv = \frac{nh}{2\pi \mathrm{r}} = \frac{3h}{2\pi \times 9a_0} = \frac{h}{6\pi a_0}$

$\lambda = \frac{h}{mv} = \frac{n}{h} \times 6\pi a_0 = 6\pi a_0$

Q.20 Answer:

half

Solution:

de − Broglie wavelength is given by :

$\lambda = \frac{h}{mv}$

$\mathrm{K.E.} = \frac{1}{2}mv^2$

$v^2 = \frac{2KE}{m}$

$v = \sqrt{\frac{2KE}{m}}$

Substituting this in equation (i) $\lambda = \frac{h}{m}\sqrt{\frac{m}{2KE}}$

$\lambda = h\sqrt{\frac{1}{2m(K.E.)}}$

i.e. $\lambda \propto \frac{1}{\sqrt{KE}}$

$\therefore$ **when** KE **become 4 times wavelength become** $1/2$**.**

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